DISCRETE STRUCTURES

CMSC250

Chapter 18: Proving Statements About Sets

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Element chasing, identities, and counterexamples



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1 The central proof strategy: element chasing

Set identities are statements about membership. To prove $A \subseteq B$, start with an *arbitrary* element $x \in A$, unpack the definition of membership, and show $x \in B$. To prove $A = B$, prove both inclusions. This is called **element chasing**.

Key idea. Translate set operations into logic: $x \in A \cup B \iff (x \in A \vee x \in B)$; $x \in A \cap B \iff (x \in A \wedge x \in B)$; $x \in A \setminus B \iff (x \in A \wedge x \notin B)$.

2 A complete distributive-law proof

Worked example: intersection distributes over union

Question. For all sets A, B, C, prove $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

Proof. We prove both inclusions. Let $x \in A \cap (B \cup C)$ be arbitrary. Then $x \in A$ and $x \in B \cup C$. Thus $x \in B$ or $x \in C$. If $x \in B$, then $x \in A \cap B$. If $x \in C$, then $x \in A \cap C$. Either way $x \in (A \cap B) \cup (A \cap C)$. This proves the first inclusion. Conversely let $x \in (A \cap B) \cup (A \cap C)$ be arbitrary. Then $x \in A \cap B$ or $x \in A \cap C$. In either case $x \in A$ and at least one of $x \in B$ or $x \in C$ holds. Therefore $x \in A \cap (B \cup C)$. Both inclusions hold, so the sets are equal. ■

Why the proof works. The proof does not assume anything special about A, B, C or the chosen element. The arbitrary-element language is what makes it a universal proof.

3 De Morgan's laws for sets

Worked example: complement of a union

Question. Given a universal set U, prove $(A \cup B)^c = A^c \cap B^c$.

Proof. Let $x \in U$ be arbitrary. Then

$$\begin{aligned} x \in (A \cup B)^c &\iff x \notin A \cup B \\ &\iff \sim (x \in A \vee x \in B) \\ &\iff (x \notin A) \wedge (x \notin B) \\ &\iff x \in A^c \cap B^c. \end{aligned}$$

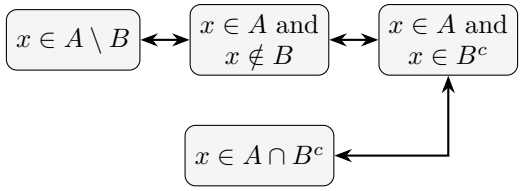
The middle step is De Morgan's law for propositions. Since membership is equivalent for every $x \in U$, the two sets are equal. ■

Worked example: complement of an intersection

Question. Prove $(A \cap B)^c = A^c \cup B^c$.

Proof. For arbitrary $x \in U$, $x \in (A \cap B)^c$ iff $x \notin A \cap B$, iff it is not the case that $x \in A$ and $x \in B$, iff $x \notin A$ or $x \notin B$, iff $x \in A^c \cup B^c$. Thus the sets are equal. ■

4 A membership-flow diagram



5 Set difference identities

Worked example: difference equals intersection with complement

Question. Prove $A \setminus B = A \cap B^c$.

Proof. Let $x \in U$ be arbitrary. By definition, $x \in A \setminus B$ iff $x \in A$ and $x \notin B$. The second condition is equivalent to $x \in B^c$. Therefore $x \in A \setminus B$ iff $x \in A \cap B^c$. Equality follows. ■

Worked example: difference distributes over union

Question. Prove $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$.

Proof. Let x be arbitrary. Then

$$\begin{aligned} x \in A \setminus (B \cup C) &\iff x \in A \wedge x \notin (B \cup C) \\ &\iff x \in A \wedge x \notin B \wedge x \notin C \\ &\iff (x \in A \wedge x \notin B) \wedge (x \in A \wedge x \notin C) \\ &\iff x \in (A \setminus B) \cap (A \setminus C). \end{aligned}$$

Since the equivalence holds for arbitrary x, the sets are equal. ■

6 Subsets and intersections

Worked example: intersection is contained in both sets

Question. Prove $A \cap B \subseteq A$.

Proof. Let $x \in A \cap B$ be arbitrary. Then by definition $x \in A$ and $x \in B$. In particular $x \in A$. Therefore $A \cap B \subseteq A$. ■

Worked example: a conditional equality

Question. Prove $A \subseteq B$ if and only if $A \cap B = A$.

Proof. ($\Rightarrow$) Assume $A \subseteq B$. Every $x \in A$ also belongs to B, so $x \in A \cap B$ and $A \subseteq A \cap B$. Conversely $A \cap B \subseteq A$ always holds. Thus equality follows. ($\Leftarrow$) Assume $A \cap B = A$. Let $x \in A$ be arbitrary. Then $x \in A \cap B$, hence $x \in B$. Thus $A \subseteq B$. ■

7 Proving an assertion false: counterexamples

A universal identity can be disproved with one choice of sets for which the sides differ. A useful counterexample is small and explicit.

Worked example: difference is not commutative

Question. Is $A \setminus B = B \setminus A$ true for all sets?

Solution. No. Take $A = \{1\}$ and $B = \varnothing$. Then $A \setminus B = \{1\}$, but $B \setminus A = \varnothing$. The proposed universal identity is false.

Worked example: an invalid distributive identity

Question. Is $A \cup (B \cap C) = (A \cup B) \cap C$ always true?

Solution. No. Let $A = \{1\}$ and $B = C = \varnothing$. The left side is $\{1\}$ and the right side is $\varnothing$. One counterexample suffices.

8 How to write polished set proofs

State the claim, declare an arbitrary element, explicitly unpack definitions, justify each implication or equivalence, and conclude the desired inclusion or equality. End a completed proof with ■. Avoid phrases such as "obviously" when a definition would give a precise reason.

9 Practice questions with complete solutions

Question. 1. Prove $A \cup (A \cap B) = A$.

Solution. Let x be arbitrary. If $x \in A \cup (A \cap B)$, then either $x \in A$ or $x \in A \cap B$, and both cases give $x \in A$. Conversely if $x \in A$, then $x \in A \cup (A \cap B)$ by definition of union. Thus the sets are equal. ■**Question.** 2. Prove $A \subseteq B$ implies $A \cup C \subseteq B \cup C$.

Solution. Assume $A \subseteq B$. Let $x \in A \cup C$ be arbitrary. If $x \in A$, the assumption gives $x \in B$ and hence $x \in B \cup C$. If $x \in C$, then $x \in B \cup C$ immediately. Therefore $A \cup C \subseteq B \cup C$. ■**Question.** 3. Prove $A \cap (B \setminus C) = (A \cap B) \setminus C$.

Solution. For arbitrary x, membership in the left side means $x \in A$, $x \in B$, and $x \notin C$. This is equivalent to $x \in A \cap B$ and $x \notin C$, which is membership in the right side. Hence equality holds. ■

10 Common mistakes

Never infer equality from one inclusion. Never replace *or* with *and* when expanding a union. When proving a conditional set claim, explicitly state the assumption before using it. A Venn diagram may suggest a claim, but element chasing supplies the formal proof.